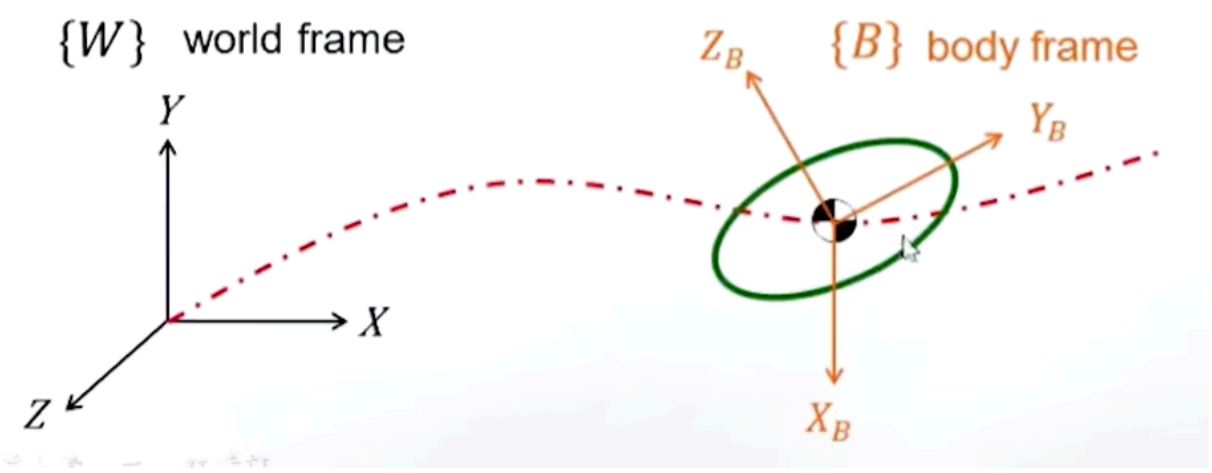
一、导论

1.1 刚体运动状态描述

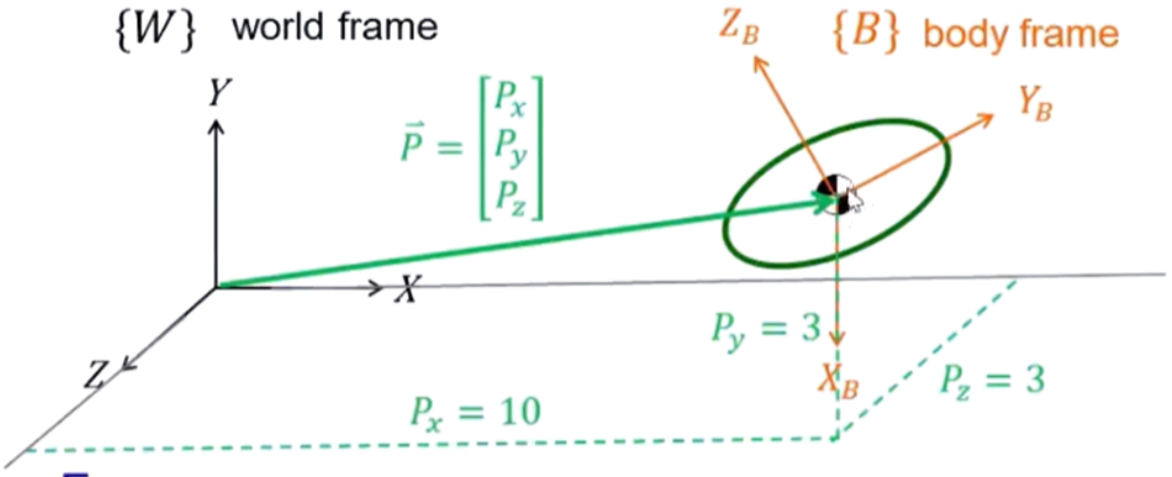
🔴该如何整合表达刚体的状态?

- 在刚体（Rigid body）上建立 frame，常建立在质心上
  - 移动：由 body frame 的**原点位置**判定
  - 转动：由 body frame 的**姿态**判定



🔴一个刚体（Rigid body）的「运动」状态该如何描述?

- 利用各个 DOF（自由度）的微分，将位移和姿态（displacement /orientation）
- 转换到速度（velocity）和加速度（acceleration）等运动状态
- 移动：以向量（vector）P 来描述{B} 的原点相对于{A} 的状态
-

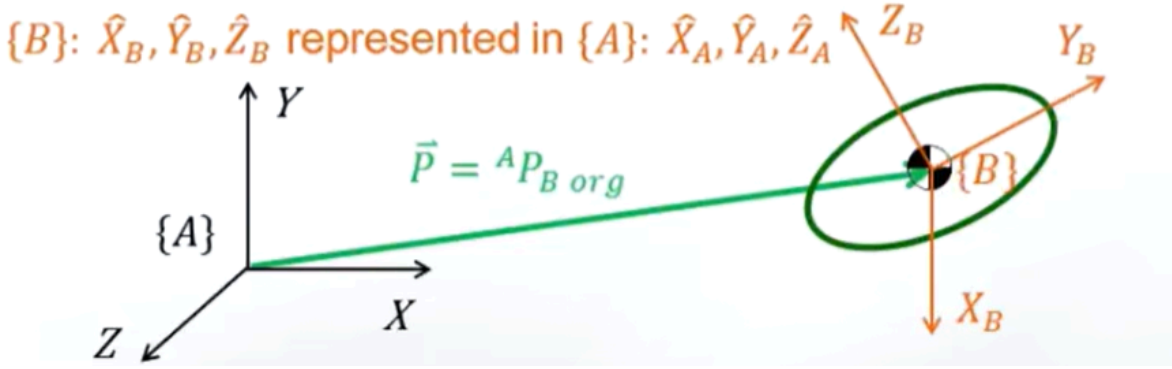


- Ex:

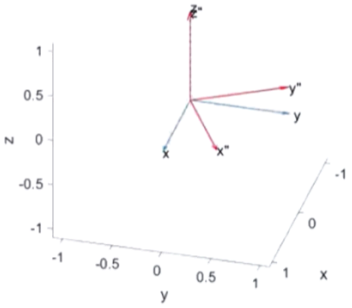
$$\vec{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$$

🔴向量可表达空间关系的两种方式：

- A position in space (i.e., position vector)如同前一页内容，以此方式描述 body frame 原点P=PxPyPz=APB org=origin of {B} represented in {A}
- A vector (i.e., displacement, frame basis)以此方式表达 body frame 上 principal axes 的方向

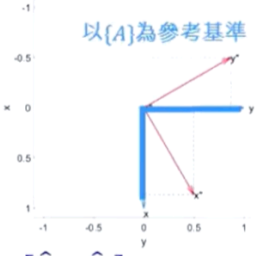


- Ex: $\{B\}$相對於$\{A\}$之姿態 ${}^{A}_{B}R$ =?



上視圖

以$\{A\}$為參考基準



藍虛線: World Frame $\{A\}$

紅實線: Body Frame $\{B\}$

$$ {}^{A}\hat{X}_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0.866 \\ 0.5 \\ 0 \end{bmatrix} $$

$$ {}^{A}\hat{Y}_B = \begin{bmatrix} \hat{Y}_B \cdot \hat{X}_A \\ \hat{Y}_B \cdot \hat{Y}_A \\ \hat{Y}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.866 \\ 0 \end{bmatrix} $$

$$ {}^{A}\hat{Z}_B = \begin{bmatrix} \hat{Z}_B \cdot \hat{X}_A \\ \hat{Z}_B \cdot \hat{Y}_A \\ \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} $$

$\{B\}$相對於$\{A\}$之姿態：

$$ {}^{A}_{B}R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$